



Islamic University of Technology
Department of Computer Science and Engineering

Lab 4: Shortest Path Algorithm

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

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Task A. Double Collection

Time Limit: 2 seconds | Memory Limit: 256 MB

A courier starts at city S and must deliver a package to city D . Before reaching D , the courier must collect two items: one from city A and one from city B . The items may be collected in **either order** — the courier chooses whichever sequence results in the shortest total journey.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of x_i . The courier always takes the fastest available route between any two cities.

Find the minimum total travel time for the courier to travel from city S to city D after collecting items from both city A and city B .

Input Format

The first line contains six integers N, M, S, D, A and B ($2 \leq N \leq 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D, A, B \leq N$). It is guaranteed that S, D, A, B are all distinct.

Each of the next M lines contains three integers u_i, v_i and x_i ($1 \leq u_i, v_i \leq N, u_i \neq v_i, 1 \leq x_i \leq 10^9$) describing a bidirectional road.

It is guaranteed that a valid route always exists. There may be multiple roads between the same pair of cities.

Output Format

Output a single integer — the minimum total travel time from city S to city D after collecting items from both city A and city B .

Examples

Sample Input	Sample Output
5 6 1 5 2 4 1 3 1 3 2 1 2 4 5 3 4 2 4 5 2 1 2 6	7
5 5 1 5 3 2 1 2 1 2 3 5 1 3 2 3 4 1 4 5 3	8

Notes

In the first sample, $S = 1, D = 5, A = 2, B = 4$. Collecting A first gives the route $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$ with distances $2 + 3 + 2 = 7$. Collecting B first gives $1 \rightarrow 4 \rightarrow 2 \rightarrow 5$ with distances $3 + 3 + 5 = 11$. The minimum is 7.

In the second sample, $S = 1$, $D = 5$, $A = 3$, $B = 2$. Collecting A first gives $1 \rightarrow 3 \rightarrow 2 \rightarrow 5$ with distances $2 + 3 + 7 = 12$. Collecting B first gives $1 \rightarrow 2 \rightarrow 3 \rightarrow 5$ with distances $1 + 3 + 4 = 8$. The minimum is 8.

Task B. Cheapest Fastest Route

Time Limit: 2 seconds | Memory Limit: 256 MB

A traveler wants to go from city 1 to city N as quickly as possible. However, roads have toll fees in addition to travel times, and the traveler wants to minimise the total toll paid — but only among routes that achieve the shortest possible travel time.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of t_i and a toll fee of c_i .

Find the minimum total toll fee among all shortest-time paths from city 1 to city N .

Input Format

The first line contains two integers N and M ($2 \leq N \leq 10^5$, $1 \leq M \leq 3 \times 10^5$).

Each of the next M lines contains four integers u_i , v_i , t_i and c_i ($1 \leq u_i, v_i \leq N$, $u_i \neq v_i$, $1 \leq t_i$, $c_i \leq 10^9$) describing a bidirectional road with travel time t_i and toll fee c_i .

It is guaranteed that city N is reachable from city 1. There may be multiple roads between the same pair of cities.

Output Format

Output a single integer — the minimum total toll fee on any shortest-time path from city 1 to city N .

Examples

Sample Input	Sample Output
4 5 1 2 3 2 1 3 2 5 2 4 2 1 3 4 3 4 2 3 1 3	3
5 6 1 2 1 5 1 3 2 1 2 4 2 3 3 4 1 4 4 5 1 2 2 5 4 1	7

Notes

In the first sample, two paths achieve the shortest travel time of 5: path $1 \rightarrow 2 \rightarrow 4$ with toll $2 + 1 = 3$, and path $1 \rightarrow 3 \rightarrow 4$ with toll $5 + 4 = 9$. The minimum toll is 3.

In the second sample, two paths achieve the shortest travel time of 4: path $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$ with toll $5 + 3 + 2 = 10$, and path $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$ with toll $1 + 4 + 2 = 7$. The minimum toll is 7.

Marks Distribution

Task	Marks
Task A	50%
Task B	50%

Practice Problems

Problem	Links
Shortest Routes I	CSES: https://cses.fi/problemset/task/1671
Jzzhu and Train Routes	Codeforces: https://codeforces.com/problemset/problem/449/B
Dijkstra?	Codeforces: https://codeforces.com/problemset/problem/20/C
Sending Email	UVA: https://onlinejudge.org/index.php?option=onlinejudge&Itemid=8&